

OS JACKFRUIT

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1.Screenshot 1 — Multi-container supervision

Description:

Two containers (alpha, beta) running simultaneously under one supervisor process

TERMINAL-1

```
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor rootfs-base  
Supervisor ready on /tmp/mini_runtime.sock
```

TERMINAL -2

```
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha rootfs-base /bin/sh  
sudo ./engine start beta rootfs-base /bin/sh  
[sudo] password for heemadhawala:  
started pid 6035  
started pid 6041
```

COMMANDS sudo ./engine supervisor rootfs-base
(TERMINAL -1) sudo ./engine start alpha rootfs-base /bin/sh
(TERMINAL-2) sudo ./engine start beta rootfs-base
/bin/sh(TERMINAL-2)

2.Screenshot 2 — Metadata tracking

Description:

Output of **engine ps** showing container ID, PID, state, and memory limits.

```
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo ./engine ps  
Expected states include: starting, running, stopped, killed, exited  
ID          PID      STATE    SOFT_MIB  HARD_MIB  
beta        6041     running  40         64  
alpha       6035     running  40         64
```

COMMANDS

sudo ./engine ps

3.Screenshot 3 — Bounded-buffer logging

Description:

Log output captured through the producer–consumer logging pipeline.

```
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo ./engine start log1 rootfs-base /bin/date
sudo ./engine logs log1
started pid 6217
Tue Apr 14 11:13:09 UTC 2026
```

COMMANDS

```
sudo ./engine start log1 rootfs-base /bin/date sudo
./engine logs log1
```

4.Screenshot 4 — CLI and IPC

Description:

A CLI **stop** command sent to the supervisor via UNIX socket and handled correctly.

```
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo ./engine start t2 rootfs-base /bin/sh
started pid 6359
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo ./engine stop t2
sent SIGTERM to 6359
```

COMMANDS

```
sudo ./engine start t2 rootfs-base /bin/sh sudo
./engine stop t2
```

5.Screenshot 5 — Soft-limit warning

Description:

Kernel logs showing warning when container exceeds soft memory limit.

```
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo ./engine start soft rootfs-base ./memory_hog
started pid 6590
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo dmesg | grep container_monitor
```

```
[ 2292.627616] [container_monitor] Registering container soft pid=6590 soft limit=67108864
[ 2296.627629] [container_monitor] SOFT LIMIT container=soft pid=6590 rss=4243
```

COMMANDS

```
sudo ./engine start soft rootfs-base ./memory_hog sudo
dmesg | grep container_monitor
```

6.Screenshot 6 — Hard-limit enforcement

Description:

Kernel logs showing process termination when hard memory limit is exceeded.

```
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo dmesg | grep container_monitor
```

```
[ 2299.702452] [container_monitor] HARD LIMIT container=soft pid=6590 rss=67624960 limit=67108864
```

7.Screenshot 7 — Scheduling experiment

Description:

Two containers running CPU workloads with different priorities showing scheduling behavior.

```

heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo ./engine start high rootfs-base ./cpu_hog --nice -5
sudo ./engine start low rootfs-base ./cpu_hog --nice 10
started pid 6945
started pid 6951
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo ./engine logs high
sudo ./engine logs low
cpu_hog alive elapsed=1 accumulator=11743365965695951113
cpu_hog alive elapsed=2 accumulator=12705706947733282540
cpu_hog alive elapsed=3 accumulator=6536592348963765490
cpu_hog alive elapsed=4 accumulator=2748803756655716212
cpu_hog alive elapsed=5 accumulator=5471689385478181812
cpu_hog alive elapsed=6 accumulator=1879574134951606724
cpu_hog alive elapsed=7 accumulator=16439616552241313599
cpu_hog alive elapsed=8 accumulator=14985577974824870241
cpu_hog alive elapsed=9 accumulator=11701429646925013122
cpu_hog done duration=10 accumulator=12397476453989931003
cpu_hog alive elapsed=1 accumulator=3853890695891366752
cpu_hog alive elapsed=2 accumulator=2052502631246739177
cpu_hog alive elapsed=3 accumulator=3747542254937611419
cpu_hog alive elapsed=4 accumulator=3569131114005427723
cpu_hog alive elapsed=5 accumulator=14262534237190810810
cpu_hog alive elapsed=6 accumulator=13503008120158151363
cpu_hog alive elapsed=7 accumulator=15226199755963252479
cpu_hog alive elapsed=8 accumulator=1200427456992044821
cpu_hog alive elapsed=9 accumulator=10509867372172427228
cpu_hog alive elapsed=10 accumulator=14679383370081987508
cpu_hog done duration=10 accumulator=14679383370081987508

```

COMMANDS

```

sudo ./engine start high rootfs-base ./cpu_hog --nice -5
sudo ./engine start low rootfs-base ./cpu_hog --nice 10
sudo ./engine logs high sudo ./engine logs low

```

8.Screenshot 8 — Clean teardown

Description:

All processes stopped and kernel module unloaded cleanly.

```

heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo pkill engine
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ ps aux | grep engine
heemadh+ 7407 0.0 0.0 6252 2008 pts/3 S+ 17:01 0:00 grep engine
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ grep engine
I

^C
heemadhawala@debian:~/OS-Jackfruit/boilerplate$ sudo rmdir monitor
sudo dmesg | tail
[ 2231.483741] [container_monitor] Registering container=soft pid=6502 soft=41943040 hard=67108864
[ 2292.027775] [container_monitor] IOCTL HIT
[ 2292.027810] [container_monitor] Registering container=soft pid=6590 soft=41943040 hard=67108864
[ 2296.627629] [container_monitor] SOFT LIMIT container=soft pid=6590 rss=42434560 limit=41943040
[ 2299.702452] [container_monitor] HARD LIMIT container=soft pid=6590 rss=67624960 limit=67108864
[ 2627.954459] [container_monitor] IOCTL HIT
[ 2627.954482] [container_monitor] Registering container=high pid=6945 soft=41943040 hard=67108864
[ 2627.978245] [container_monitor] IOCTL HIT
[ 2627.978249] [container_monitor] Registering container=low pid=6951 soft=41943040 hard=67108864
[ 3219.004801] [container_monitor] Module unloaded.

```

COMMANDS

```

sudo pkill engine ps
aux | grep engine
sudo rmdir
monitor sudo dmesg
| tail

```